**Chemistry • Formulas, moles and reacting masses**

# Stoichiometry

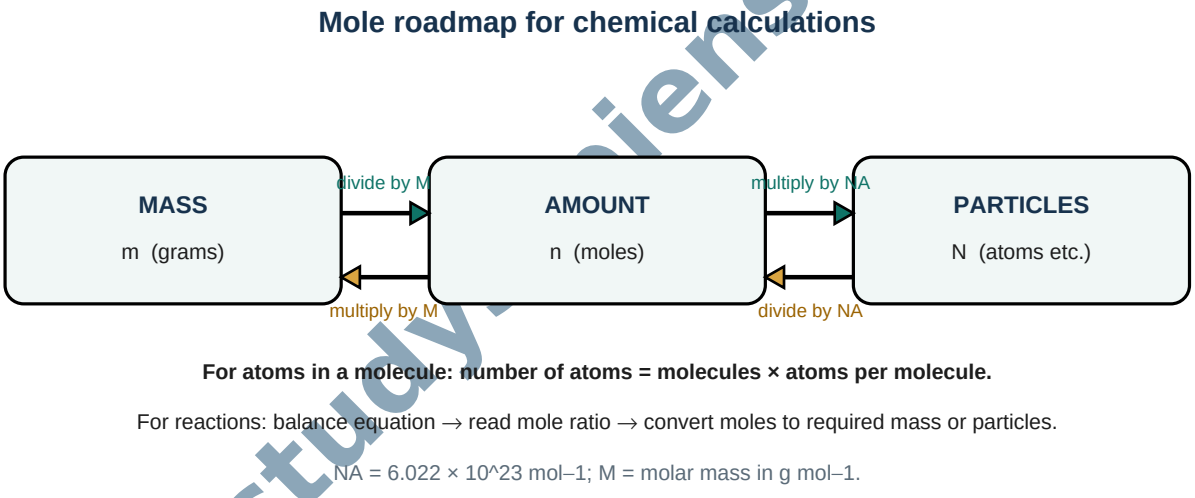
Official exercise • Complete solved answers • Fully worked numericals

**Chapter 4 • Complete official exercise • 10 MCQs + 5 short + 5 constructed + 7 descriptive + 2 investigative**

## What this chapter is about

• **Empirical formula** gives the simplest whole-number atom ratio; **molecular formula** gives the actual atom count in a molecule.
• **One mole** contains $6.022 \times 10^{23}$ specified particles. Molar mass links laboratory mass to amount of substance.
• **A balanced chemical equation** supplies mole ratios. Convert the known substance to moles, apply the coefficient ratio, then convert to the requested unit.
• **Percentage composition** compares each element's mass with the total compound mass; purity corrections adjust the mass of an impure sample.

**Figure 1 — Original roadmap for mass–mole–particle and reaction calculations**




**Exercise coverage:** All five official sections are solved in book order. Scientific notation, reaction coefficients and numerical options were checked against the rendered exercise.

## A. Multiple Choice Questions

### 1. How many atoms are present in one gram of $H_2O$?

(a) $1002 \times 10^{23}$ atoms
(b) $6.022 \times 10^{23}$ atoms
(c) $0.334 \times 10^{23}$ atoms
(d) $2.004 \times 10^{23}$ atoms

**Answer: (a) $1002 \times 10^{23}$ atoms** — $n(H_2O) = 1/18$ mol; atoms = $(1/18) \times 6.022 \times 10^{23}$ molecules $\times 3 = 1.0037 \times 10^{23}$. Option (a) is intended: its printed "1002" evidently omits the decimal in about $1.002 \times 10^{23}$.

**2. Which is the correct formula of calcium phosphide?**

(a) CaP
(b) $CaP_2$
(c) $Ca_2P_3$
(d) $Ca_3P_2$

**Answer: (d) $Ca_3P_2$** — $Ca^{2+}$ and $P^{3-}$ combine in a 3:2 ratio, giving zero net charge: $3(+2) + 2(-3) = 0$.

**3. How many atomic mass units (amu) are there in one gram?**

(a) 1 amu
(b) $10^{23}$ amu
(c) $6.022 \times 10^{23}$ amu
(d) $6.022 \times 10^{22}$ amu

**Answer: (c) $6.022 \times 10^{23}$ amu** — By definition, 1 amu ≈ $1/(6.022 \times 10^{23})$ g, so 1 g = $6.022 \times 10^{23}$ amu.

**4. The structural formula of 2-hexene is $CH_3$-CH=CH-$(CH_2)_2$-$CH_3$. What will be its empirical formula?**

(a) $C_2H_2$
(b) CH
(c) $C_6H_{12}$
(d) $CH_2$

**Answer: (d) $CH_2$** — The molecular formula is $C_6H_{12}$. Dividing the subscripts by 6 gives the simplest ratio $CH_2$.

**5. How many moles are there in 25 g of $H_2SO_4$?**

(a) 0.765 moles
(b) 0.51 moles
(c) 0.255 moles
(d) 0.4 moles

**Answer: (c) 0.255 moles** — $M(H_2SO_4) = 98$ g mol$^{-1}$, so n = 25/98 = 0.255 mol.

**6. A necklace has 6 g of diamonds in it. What is the number of carbon atoms in it?**

(a) $6.02 \times 10^{23}$
(b) $12.04 \times 10^{23}$
(c) $1.003 \times 10^{23}$
(d) $3.01 \times 10^{23}$

**Answer: (d) $3.01 \times 10^{23}$** — Diamond is carbon: n = 6/12 = 0.5 mol, so N = $0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$ atoms.

**7. What is the mass of Al in 204 g of aluminium oxide, $Al_2O_3$?**

(a) 26 g
(b) 27 g
(c) 54 g
(d) 108 g

**Answer: (d) 108 g** — $M(Al_2O_3) = 102$ g mol$^{-1}$; 204 g = 2 mol. These contain 4 mol Al, with mass $4 \times 27 = 108$ g.

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**8. Which one of the following compounds will have the highest percentage of the mass of nitrogen?**

(a) $CO(NH_2)_2$ (b) $N_2H_4$
(c) $NH_3$ (d) $NH_2OH$

**Answer: (b) $N_2H_4$** — N percentages are approximately 46.7%, 87.5%, 82.4% and 42.4% respectively, so $N_2H_4$ is highest.

**9. When one mole of each of the following compounds is reacted with oxygen, which will produce the maximum amount of $CO_2$?**

(a) carbon (b) diamond
(c) ethane ($C_2H_6$) (d) methane ($CH_4$)

**Answer: (c) ethane ($C_2H_6$)** — One mole of ethane contains 2 mol C and produces 2 mol $CO_2$; one mole of C, diamond carbon or $CH_4$ produces 1 mol $CO_2$.

**10. What mass of 95% $CaCO_3$ will be required to neutralize 50 $cm^3$ of 0.5 M HCl solution?**

(a) 9.5 g (b) 1.25 g
(c) 1.32 g (d) 1.45 g

**Answer: (c) 1.32 g** — n(HCl) = 0.050 × 0.5 = 0.025 mol; n($CaCO_3$) = 0.0125 mol. Pure $CaCO_3$ = 1.25 g; 95% sample = 1.25/0.95 = 1.316 ≈ 1.32 g.

## B. Questions for Short Answers

**4.1. Write down the chemical formula of barium nitride.**

Barium forms $Ba^{2+}$ and nitrogen forms $N^{3-}$. The neutral 3:2 combination is **$Ba_3N_2$**: 3(+2) + 2(−3) = 0.

**4.2 Find the molecular formula of a compound whose empirical formula is $CH_2O$ and whose molar mass is 180 g $mol^{-1}$.**

**Given:** Empirical formula = $CH_2O$; molar mass = 180 g $mol^{-1}$.

Empirical formula mass = 12 + 2(1) + 16 = 30 g $mol^{-1}$

n = molar mass / empirical formula mass = 180/30 = 6

Molecular formula = $(CH_2O)_6$

**Answer: $C_6H_{12}O_6$**

**4.3 How many molecules are present in 1.5 g $H_2O$?**

**Given:** m = 1.5 g; M($H_2O$) = 18 g $mol^{-1}$; $N_A$ = 6.022 × $10^{23}$ $mol^{-1}$.

n = m/M = 1.5/18 = 0.08333 mol

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$$N = nN_A = 0.08333 \times 6.022 \times 10^{23}$$

**Answer: $5.018 \times 10^{22}$ $H_2O$ molecules**

### 4.4. What is the difference between a mole and Avogadro's number?

A **mole** is the SI unit for amount of substance: one mole contains exactly $6.02214076 \times 10^{23}$ specified entities. **Avogadro's number** is that fixed number of entities, usually written $6.022 \times 10^{23}$; the Avogadro constant has unit $mol^{-1}$. Thus mole is an amount/unit, whereas Avogadro's number is the particle count associated with one mole.

### 4.5. Write the chemical equation for: Copper + sulphuric acid → copper sulphate + sulphur dioxide + water.

$$Cu(s) + 2H_2SO_4(conc., aq) \rightarrow CuSO_4(aq) + SO_2(g) + 2H_2O(l)$$

The equation is balanced: Cu = 1, H = 4, S = 2 and O = 8 on each side. Hot concentrated sulphuric acid is required for this reaction; dilute sulphuric acid does not normally react with copper in this way.

## C. Constructed Response Questions

### 4.1. "Different compounds will never have the same molecular formula but they can have the same empirical formula." Explain.

**Correction:** The first clause is too absolute. Different compounds **can** have the same molecular formula; such compounds are **isomers**. For example, ethanol ($CH_3CH_2OH$) and dimethyl ether ($CH_3OCH_3$) both have molecular formula $C_2H_6O$ but different structures and properties.

The second clause is correct. An empirical formula records only the simplest atom ratio, so different molecular formulas may reduce to the same empirical formula. Ethene $C_2H_4$ and propene $C_3H_6$ both have empirical formula $CH_2$. Likewise, $NO_2$ and $N_2O_4$ have the same empirical formula $NO_2$.

Therefore, a molecular formula alone does not always identify a unique compound; structure may also be required.

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### 4.2. Write down the chemical formulas of calcium phosphate, aluminium nitride, sodium acetate, ammonium carbonate and bismuth sulphate.

| Compound | Ions / group | Formula |
|---|---|---|
| Calcium phosphate | $Ca^{2+}$, $PO_4^{3-}$ | $Ca_3(PO_4)_2$ |
| Aluminium nitride | $Al^{3+}$, $N^{3-}$ | AlN |
| Sodium acetate | $Na^{+}$, $CH_3COO^{-}$ | $CH_3COONa$ |
| Ammonium carbonate | $NH_4^{+}$, $CO_3^{2-}$ | $(NH_4)_2CO_3$ |
| Bismuth sulphate | $Bi^{3+}$, $SO_4^{2-}$ | $Bi_2(SO_4)_3$ |

Each formula has zero total ionic charge. Parentheses show that the entire polyatomic ion is multiplied by the following subscript.

### 4.3. Why does Avogadro’s number have immense importance in chemistry?

Atoms and molecules are far too small to count individually in an ordinary sample. Avogadro’s number links that microscopic particle count to a measurable macroscopic amount: one mole. Together with molar mass it lets us convert grams to numbers of atoms, ions or molecules, use balanced-equation mole ratios, prepare concentrations and compare reacting quantities quantitatively.

### 4.4 When 8.657 g of a compound was converted into elements, it gave 5.217 g carbon, 0.962 g hydrogen and 2.478 g oxygen. Calculate the percentage of each element.

**Given:** Total mass = 8.657 g; C = 5.217 g; H = 0.962 g; O = 2.478 g. Check: 5.217 + 0.962 + 2.478 = 8.657 g.

$$\%C = (5.217/8.657) \times 100 = 60.2634\%$$

$$\%H = (0.962/8.657) \times 100 = 11.1124\%$$

$$\%O = (2.478/8.657) \times 100 = 28.6242\%$$

**Answer: C = 60.26%, H = 11.11%, O = 28.62% (total 99.99% after rounding)**

### 4.5. How can you calculate the masses of the products formed in a reversible reaction?

A reversible reaction reaches equilibrium with reactants and products present, so complete conversion cannot normally be assumed.

**1.** Write the balanced equation and set out initial amounts and the change in moles using its coefficients.

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**2.** Determine the equilibrium amounts from measured equilibrium concentrations or from K and the initial conditions. If percentage yield is supplied, the equilibrium extent may instead be inferred from that yield.

**3.** Convert each equilibrium amount to mass with m = nM. For a stated percentage yield, actual product mass = theoretical mass × yield/100.

$$\text{For } aA \rightleftharpoons bB\text{: if extent is } \xi, \text{ then } n(B) = n_0(B) + b\xi \text{ and } n(A) = n_0(A) - a\xi$$

$$\text{Product mass } m(B) = n(B) \times M(B)$$

The stoichiometric ratio gives the **change** in amounts; the equilibrium data determine how far the reaction actually proceeds.

## D. Descriptive Questions

### 4.1. Which conditions must be fulfilled before writing a chemical equation for a reaction?

**1. Identify the substances:** know the actual reactants and products from experimental evidence.

**2. Use correct formulas:** write correct element symbols, ionic ratios and molecular formulas; never change subscripts merely to balance an equation.

**3. Conserve atoms:** place coefficients so the number of atoms of every element is equal on both sides. Total charge must also be conserved in ionic equations.

**4. Use simplest coefficients:** reduce coefficients to the smallest whole-number ratio and use the correct arrow (→, ⇌) for the reaction type.

**5. Include relevant information:** physical states and essential conditions such as heat, catalyst, pressure or concentration should be shown when required.

Finally, recount every element and check that formulas—not just atom counts—represent the real substances.

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### 4.2. Explain the concepts of Avogadro's number and mole.

**Avogadro's number** is $6.022 \times 10^{23}$ (exactly $6.02214076 \times 10^{23}$ in the modern SI definition). It is the number of specified elementary entities in one mole.

**A mole** is the amount of substance containing that number of atoms, molecules, ions, electrons or other specified particles. The kind of particle must always be stated.

- 1 mol C atoms = $6.022 \times 10^{23}$ C atoms and has mass about 12 g.
- 1 mol $H_2O$ molecules = $6.022 \times 10^{23}$ molecules and has mass 18 g.
- 1 mol $H_2O$ contains 2 mol H atoms and 1 mol O atoms, hence 3 mol atoms = $3N_A$ atoms in total.

$$n = m/M \text{ and } N = nN_A$$

Molar mass M is the mass of one mole, expressed in g $mol^{-1}$. It provides the bridge between weighing a sample and counting its particles.

### 4.3 How many grams of $CO_2$ will be produced when 10 g $CH_4$ reacts with excess $O_2$? $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$.

**Given:** $M(CH_4) = 16$ g $mol^{-1}$; $M(CO_2) = 44$ g $mol^{-1}$; $O_2$ is in excess.

$$n(CH_4) = 10/16 = 0.625 \text{ mol}$$

Mole ratio $CH_4 : CO_2 = 1 : 1$, so $n(CO_2) = 0.625$ mol

$$m(CO_2) = nM = 0.625 \times 44$$

**Answer: 27.5 g $CO_2$**

### 4.4 How many moles of coal are needed to produce 10 moles of CO? $3C(s) + O_2(g) + H_2O(l) \rightarrow H_2(g) + 3CO(g)$.

**Given:** Required n(CO) = 10 mol; equation ratio C : CO = 3 : 3 = 1 : 1.

$$n(C) = (3/3) \times n(CO) = 10 \text{ mol}$$

**Answer: 10 mol of carbon represented by the coal**

Coal is a mixture, so “moles of coal” strictly means moles of carbon atoms taking part in the given idealized equation.

### 4.5 How much $SO_2$ is needed in grams to produce 10 moles of sulphur? $2H_2S(g) + SO_2(g) \rightarrow 2H_2O(l) + 3S(s)$.

**Given:** Required n(S) = 10 mol; $M(SO_2) = 32 + 2(16) = 64$ g $mol^{-1}$.

Mole ratio $SO_2 : S = 1 : 3$

$$n(SO_2) = 10/3 = 3.333 \text{ mol}$$

$$m(SO_2) = 3.333 \times 64$$

**Answer: 213.33 g $SO_2$**

### 4.6 How much ammonia is needed in grams to produce 1 kg of urea fertilizer? $2NH_3(aq) + CO_2(aq) \rightarrow CO(NH_2)_2(aq) + H_2O(l)$.

**Given:** m(urea) = 1000 g; M(urea) = 60 g $mol^{-1}$; $M(NH_3) = 17$ g $mol^{-1}$.

$$n(\text{urea}) = 1000/60 = 16.6667 \text{ mol}$$

Mole ratio $NH_3$ : urea = 2 : 1

$$n(NH_3) = 2 \times 16.6667 = 33.3333 \text{ mol}$$

$$m(NH_3) = 33.3333 \times 17$$

**Answer: 566.67 g $NH_3$**

**4.7. Calculate the number of atoms in: (a) 3 g $H_2$, (b) 3.4 moles $N_2$, (c) 10 g $C_6H_{12}O_6$.**

**(a) 3 g of hydrogen gas**

$$n(H_2) = 3/2 = 1.5 \text{ mol}; \; n(\text{H atoms}) = 2 \times 1.5 = 3 \text{ mol}$$

$$N(\text{H atoms}) = 3 \times 6.022 \times 10^{23} = \mathbf{1.8066 \times 10^{24} \text{ atoms}}$$

**(b) 3.4 mol of nitrogen gas**

$$n(\text{N atoms}) = 2 \times 3.4 = 6.8 \text{ mol}$$

$$N(\text{N atoms}) = 6.8 \times 6.022 \times 10^{23} = \mathbf{4.09496 \times 10^{24} \text{ atoms}}$$

**(c) 10 g of glucose**

$$M(C_6H_{12}O_6) = 180 \text{ g mol}^{-1}; \; n(\text{molecules}) = 10/180 = 0.05556 \text{ mol}$$

Each molecule contains 6 + 12 + 6 = 24 atoms

$$n(\text{total atoms}) = (10/180) \times 24 = 1.3333 \text{ mol}$$

$$N = 1.3333 \times 6.022 \times 10^{23} = \mathbf{8.0293 \times 10^{23} \text{ atoms}}$$

## E. Investigative Questions

**4.1 It is generally believed that drinking eight glasses of water every day is required to keep oneself hydrated, especially in summer. If a glass occupies 400 $cm^3$ of water on average, how many moles of water are needed for a single adult?**

**Given:** 8 glasses; V = 400 $cm^3$ per glass; water density ≈ 1 g $cm^{-3}$; $M(H_2O)$ = 18 g $mol^{-1}$.

$$\text{Total volume} = 8 \times 400 = 3200 \text{ cm}^3$$

$$\text{Mass of water} = 3200 \text{ cm}^3 \times 1 \text{ g cm}^{-3} = 3200 \text{ g}$$

$$n = m/M = 3200/18$$

**Answer: 177.78 mol $H_2O$ per day under the stated assumption**

Actual hydration needs vary with body size, activity, food, climate and health; the calculation addresses only the eight-glass assumption in the question.

### 4.2. The chemical formula for sand is $SiO_2$, but sand does not exist as discrete molecules like $H_2O$. How has its formula been determined keeping in view its structure?

In the quartz network, each Si atom is covalently bonded to four O atoms in a tetrahedron. Each O atom is shared as a bridge between two Si atoms.

Per silicon: 4 shared O atoms × 1/2 each = 2 O atoms

Simplest whole-number ratio Si : O = 1 : 2

Thus **$SiO_2$ is an empirical formula**, obtained from the network's atom-sharing ratio and confirmed by elemental composition. It is not the molecular formula of a separate $SiO_2$ molecule.

## Quick Revision: Molar-Mass Toolkit

**Using H = 1, C = 12, N = 14, O = 16, S = 32 and Ca = 40**

| Substance | Formula | M / g $mol^{-1}$ | Substance | Formula | M / g $mol^{-1}$ |
|---|---|---|---|---|---|
| Hydrogen | $H_2$ | 2 | Water | $H_2O$ | 18 |
| Methane | $CH_4$ | 16 | Ammonia | $NH_3$ | 17 |
| Carbon dioxide | $CO_2$ | 44 | Sulphur dioxide | $SO_2$ | 64 |
| Sulphuric acid | $H_2SO_4$ | 98 | Calcium carbonate | $CaCO_3$ | 100 |
| Urea | $CO(NH_2)_2$ | 60 | Glucose | $C_6H_{12}O_6$ | 180 |

**Fast check:** a mole calculation should end in mol, a particle count should end in atoms/molecules/ions, and a final mass should return to grams.

## Points to Remember

**Molecular formula:** actual atom counts in a molecule. **Empirical formula:** simplest whole-number atom ratio.

Molecular formula = $(\text{empirical formula})_n$, where n = molar mass ÷ empirical formula mass.

$n = m/M$ and $N = nN_A$; for compounds, multiply by atoms or ions per formula unit when required.

$N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$; molar mass has unit $\text{g mol}^{-1}$.

Ionic formulas are neutral overall; use ion charges and parentheses correctly.

Balanced-equation coefficients give mole ratios, not direct gram ratios.

Percentage by mass = (mass of element or component / total mass) × 100.

For purity: impure sample mass = required pure mass ÷ fractional purity. For reversible reactions, equilibrium extent or yield is also required.

### MCQ key at a glance

| Q | Key | Q | Key | Q | Key | Q | Key | Q | Key |
|---|---|---|---|---|---|---|---|---|---|
| 1 | a | 2 | d | 3 | c | 4 | d | 5 | c |
| 6 | d | 7 | d | 8 | b | 9 | c | 10 | c |

### How to attempt this chapter in the paper

- Write the balanced equation first and underline the coefficient ratio linking the given and requested substances.
- Keep units in every line: grams → moles → moles of product → grams or particles.
- For atom counts, distinguish molecules from atoms: $H_2$ contains two H atoms per molecule.
- Check whether percentages total 100% within rounding and whether purity or yield requires an extra conversion.